

STAGE 2 CHEMISTRY 2020
SKILLS & APPLICATION TASK 3
TOPIC 3: ORGANIC & BIOLOGICAL CHEMISTRY
SACE Number: SOLUTIONS **Initials:** _____

School Number: _____

Investigation, Analysis, and Evaluation		Knowledge and Application
A	Critically deconstructs a problem and designs a logical, coherent, and detailed chemistry investigation. Obtains, records, and represents data using appropriate conventions and formats accurately and highly effectively. Systematically analyses and interprets data and evidence to formulate logical conclusions with detailed justification. Critically and logically evaluates procedures and discusses their effect on data.	Demonstrates deep and broad knowledge and understanding of a range of chemical concepts. Develops and applies chemical concepts highly effectively in new and familiar contexts. Critically explores and understands in depth the interaction between science and society. Communicates knowledge and understanding of chemistry coherently, with highly effective use of appropriate terms, conventions, and representations.
B	Logically deconstructs a problem and designs a well-considered and clear chemistry investigation. Obtains, records, and represents data using appropriate conventions and formats mostly accurately and effectively. Logically analyses and interprets data and evidence to formulate suitable conclusions with reasonable justification. Logically evaluates procedures and their effect on data.	Demonstrates some depth and breadth of knowledge and understanding of a range of chemical concepts. Develops and applies chemical concepts mostly effectively in new and familiar contexts. Logically explores and understands in some depth the interaction between science and society. Communicates knowledge and understanding of chemistry mostly coherently, with effective use of appropriate terms, conventions, and representations.
C	Deconstructs a problem and designs a considered and generally clear chemistry investigation. Obtains, records, and represents data, using generally appropriate conventions and formats, with some errors but generally accurately and effectively. Undertakes some analysis and interpretation of data and evidence to formulate generally appropriate conclusions with some justification. Evaluates procedures and some of their effect on data.	Demonstrates knowledge and understanding of a general range of chemical concepts. Develops and applies chemical concepts generally effectively in new or familiar contexts. Explores and understands aspects of the interaction between science and society. Communicates knowledge and understanding of chemistry generally effectively, using some appropriate terms, conventions, and representations.
D	Prepares a basic deconstruction of a problem and an outline of a chemistry investigation. Obtains, records, and represents data using conventions and formats inconsistently, with occasional accuracy and effectiveness. Describes data and undertakes some basic interpretation to formulate a basic conclusion. Attempts to evaluate procedures or suggest an effect on data.	Demonstrates some basic knowledge and partial understanding of chemical concepts. Develops and applies some chemical concepts in familiar contexts. Partially explores and recognises aspects of the interaction between science and society. Communicates basic chemical information, using some appropriate terms, conventions, and/or representations.
E	Attempts a simple deconstruction of a problem and a procedure for a chemistry investigation. Attempts to record and represent some data, with limited accuracy or effectiveness. Attempts to describe results and/or interpret data to formulate a basic conclusion. Acknowledges that procedures affect data.	Demonstrates limited recognition and awareness of chemical concepts. Attempts to develop and apply chemical concepts in familiar contexts. Attempts to explore and identify an aspect of the interaction between science and society. Attempts to communicate information about chemistry.

Description of assessment

This is a summative test on Topic 3 – Organic & Biological Chemistry. It consists of a series of short answer questions and a SHE question. All questions are to be answered in the space provided.

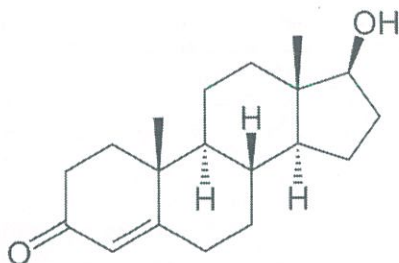
Assessment conditions

This assessment is conducted under direct supervision in a **time frame of 70 minutes**, plus **reading time of 5 minutes**. Calculators are permitted.

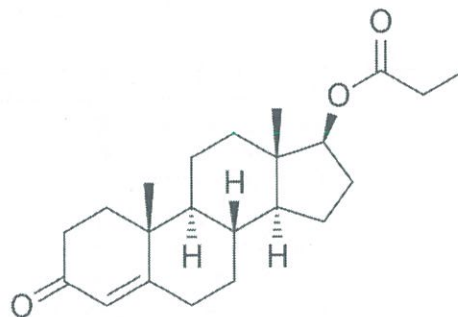
TOTAL MARKS: 67

QUESTION 1

Testosterone propanoate is a drug commonly used in the bodybuilding community to enhance muscle mass and definition. Testosterone propanoate can be prepared from testosterone by Fischer esterification.



Testosterone



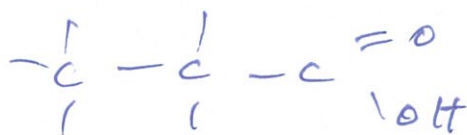
Testosterone propanoate

- (a) Name two functional groups found in the structure of testosterone propanoate.

ketone ester
alkene

[2]

- (b) Draw the structure of the second reactant needed to convert testosterone into testosterone propanoate.



[2]

- (c) Name the structure of the compound drawn in (b) above.

propanoic acid

[1]

- (d) State the reaction conditions needed to convert testosterone into testosterone propanoate.

- conc. H_2SO_4 catalyst
- heat under reflux

[2]

- (e) The drug is commonly dissolved in an oil and injected under the skin. Explain why testosterone propanoate is not very soluble in water.

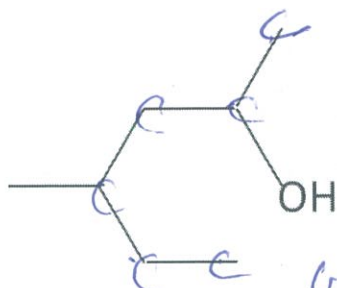
- Very large molecule
- Few sites for hydrogen bonding with polar water
- Hence only slightly soluble

[3]

QUESTION 2

Alcohols have a long history of many uses, including fuel additives, solvents, and beverages.

(a) Name **Compound A** below.



Compound A

4-methylhexan-2-ol

Name: _____ [1]

(b) State whether Compound A is a primary, secondary or tertiary alcohol.

_____ [1]

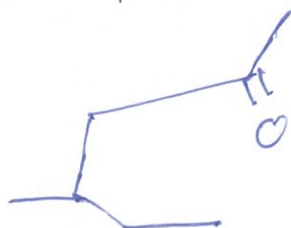
(c) Identify the colour change that occurs when Compound A reacts with excess acidified dichromate ($\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$).

orange $\rightarrow$ green

(2) ?

_____ [1]

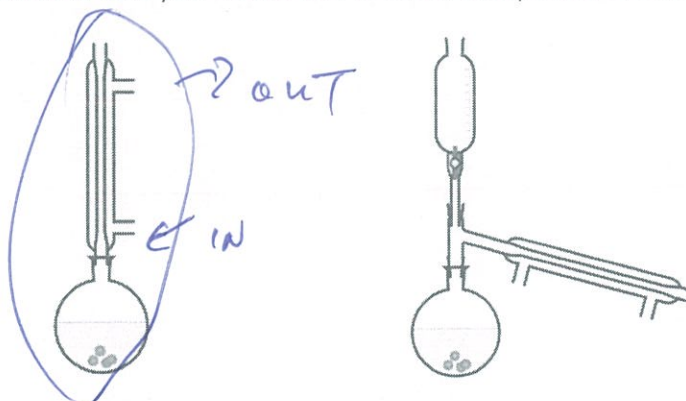
(d) Draw the structure of the compound formed when Compound A reacts with excess acidified dichromate.



[2]

(e) Compound A also reacts with ethanoic acid under reflux.

(i) Circle the apparatus that you would use to react Compound A with ethanoic acid under reflux.

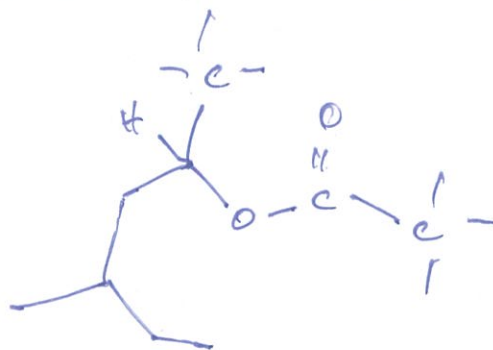


[1]

(ii) Show the direction of the water flow on your chosen apparatus.

[1]

- (iii) Draw the product formed in the reaction between Compound A and ethanoic acid.



[2]

- (iv) Name the new functional group formed when Compound A reacts with ethanoic acid.

Ester

[1]

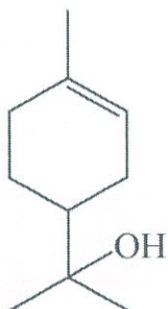
QUESTION 3

The characteristic aromas of many plants and plant products (e.g. oranges, lemons, grapes, and wines) are often caused by a group of compounds called "terpenes" and their oxygen derivatives called "terpenoids".

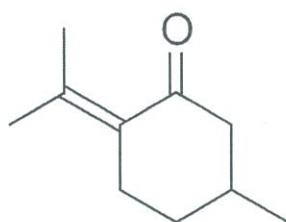
Structurally, these compounds contain ten carbon atoms per molecule. Chemically, they contain a variety of functional groups.

An unknown terpenoid is thought to be one of the four below.

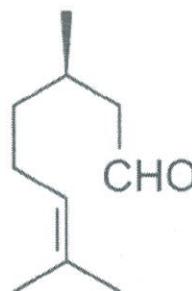
A – terpineol



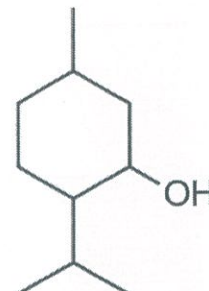
B – pulegone



C – citronellal



D – menthol



When the unknown terpenoid is subjected to two tests, it gives the following results:

Test number	Tested with	Result with unknown terpenoid
1	Bromine water	Positive
2	Tollens' reagent (ammoniacal silver nitrate)	Positive

(a) For each test, name one functional group that could be identified by using the reagents listed.

Test 1: c/c double bond / alkene [1]

Test 2: aldehyde [1]

(b) What observation would indicate a positive result in each of the tests?

Test 1 observation: decolourises immediately [1]

Test 2 observation: silver mirror forms [1]

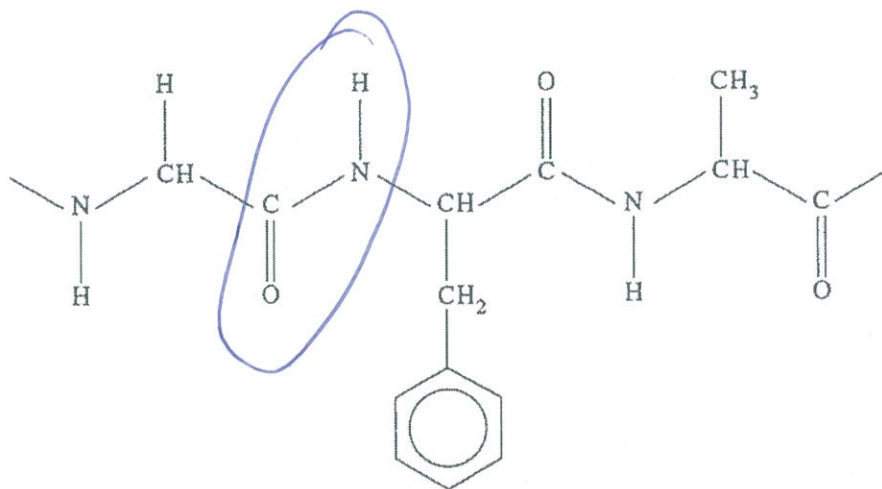
(c) Use the results in the table to identify the unknown terpenoid.

C [1]

QUESTION 4

Plants absorb nitrogen compounds and use them to produce amino acids and then proteins.

The following diagram represents a section of a protein chain.



(a) Circle a peptide group on the diagram above.

(b) State the number of amino acids bonded together in this protein section.

3

(c) Draw one of the amino acids which forms part of this protein chain.



QUESTION 5

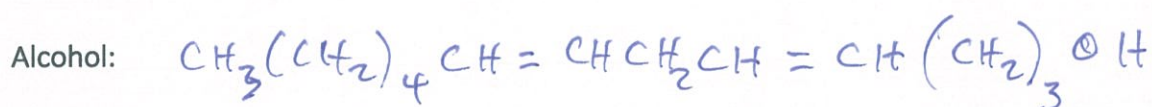
Pheromones are chemicals used by insects to communicate with other insects of the same species.

The structural formula (X) shown below is of a pheromone secreted by the potato tubeworm moth:



Compound X

- (a) Write the formula of the alcohol and the carboxylic acid from which Compound X could be prepared in a laboratory.



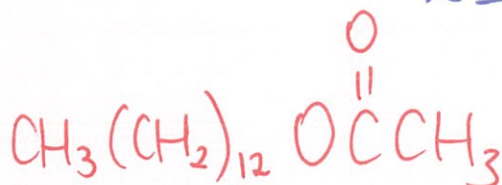
Carboxylic acid:



[2]

- (b) Compound X can be converted into a saturated compound by reacting it with hydrogen gas under pressure, in the presence of a catalyst.

- (i) Write the structural formula of the completely saturated compound formed from Compound X.



[1]

- (ii) Give a reason for the use of a catalyst in the process.

Inc. rate of reaction

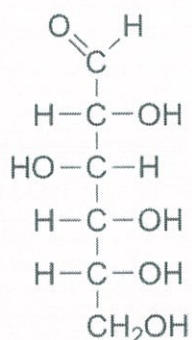
[1]

(c) Potatoes are a good source of starch, a carbohydrate consisting of many glucose units linked together.

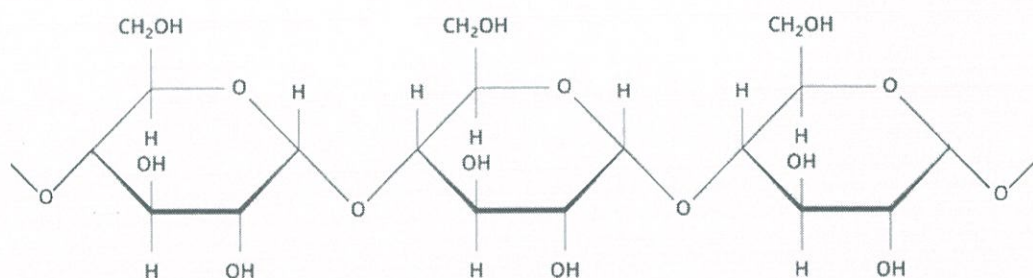
(i) State the definition of a carbohydrate.

polyhydroxy aldehyde
or
polyhydroxy ketone [2]

(ii) The diagrams below show the structures of chain glucose and a section of the starch polymer. Explain a chemical test that could be done in a laboratory to distinguish between the two. Include reagents and observations.



Glucose – chain form



Starch

-Tollen's reagent

-warm gently

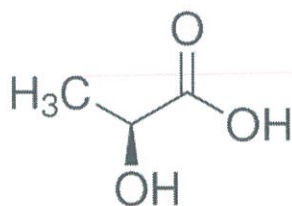
-glucose silver mirror

aldehyde gp

-starch no observable change
stays colourless

[3]

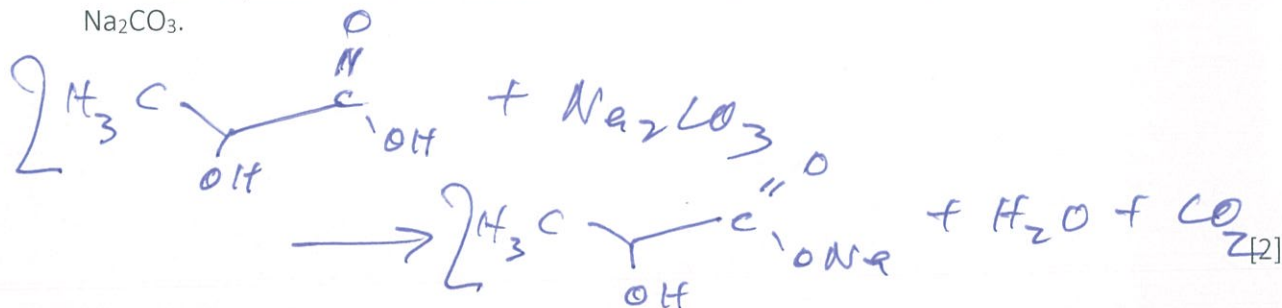
- (iii) Glucose can be converted into lactic acid. Its structural formula is below.



Lactic acid

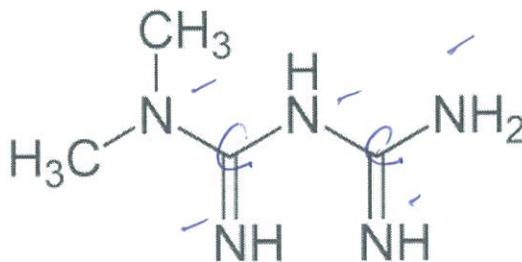
- (1) Lactic acid can be separated from a non-polar organic mixture by reacting it with aqueous sodium carbonate solution.

Write an equation for the reaction between lactic acid and aqueous sodium carbonate, Na_2CO_3 .



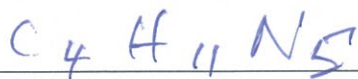
- (iv) People with diabetes have higher than normal levels of glucose in the blood. The drug *Metformin* controls blood glucose levels of people with diabetes. It can be administered in tablet form.

The chemical structure of *Metformin* is below:



Metformin

- (1) Write the molecular formula of *Metformin*.



[1]

- (2) Would you expect *Metformin* to be soluble in water? Explain.

Yes / electronegative 'N'
5 sites for H₂ bonding
reasonably soluble

[3]

- (3) A team in South Korea is developing a new way for diabetics take *Metformin*. Read the article below and answer the question that follows.

The article below comes from New Scientist magazine, Issue 3066, published 26 March 2016.

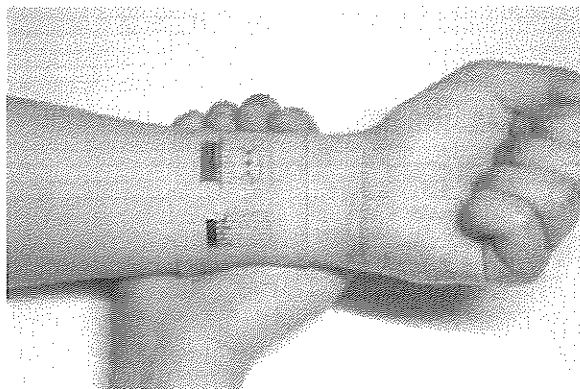
Graphene smart patch for monitoring diabetes could save lives

A non-invasive device that promises the end of finger-prick glucose tests and injections could be a boon for millions of people with diabetes.

22 March 2016

By Sam Wong

A team in South Korea... is developing a wearable device to keep diabetes under control. The patch, shown above, is made from graphene and senses glucose in sweat. It is connected by wires to a portable electrochemical analyser. When blood sugar is too high, it releases the drug metformin through microneedles to lower it.



Graphene is thin and flexible, making it comfortable to wear on the skin. When the device is worn, the microneedles pierce the skin, reaching the subcutaneous tissue. At 250 micrometres in diameter and 1 millimetre high they are too small to cause pain.

To release metformin when it is needed, the patch switches on a tiny heater embedded within. This dissolves a coating on the microneedles so that they release metformin.

Tests in diabetic mice showed that the patch is effective at sensing high glucose levels, releasing an appropriate dose of metformin and lowering blood sugar. It was also tested on two healthy men, showing that its glucose measurements are accurate and correlate well with blood concentrations (*Nature Nanotechnology*, [doi.org/bdhhb](https://doi.org/10.1038/nnano.2015.250)).

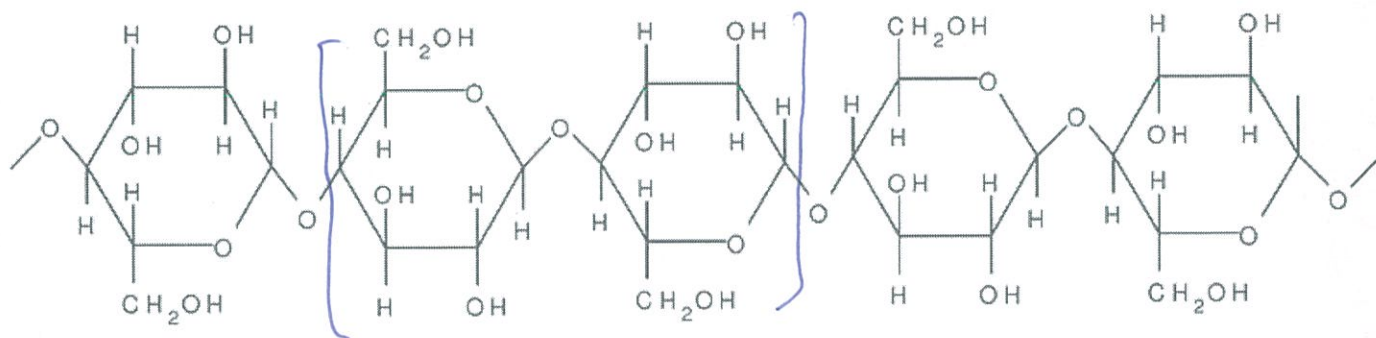
The work is still in its early stages, but the team hopes to make the patch suitable for everyday human use. “We need to reduce the cost and increase the reliability of the sensors over a longer period of time,” says Dae-Hyeong Kim at Seoul National University in South Korea.

But it’s some way from being a complete solution. One problem is that the doses of drug needed in humans are too big to be delivered by the patch, says Guy. “Even with a device that makes holes in your skin to poke it through, I have a hard time figuring out how it’s going to work... My guess is it may take another five to 10 years,” says Guy.

QUESTION 6

Cellulose is a polysaccharide that is found in the cell walls of plants.

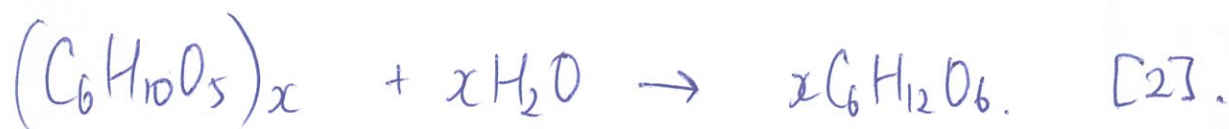
The structural formula for a short chain of cellulose is shown below.



(a) Identify the repeating unit for cellulose on the structure above.

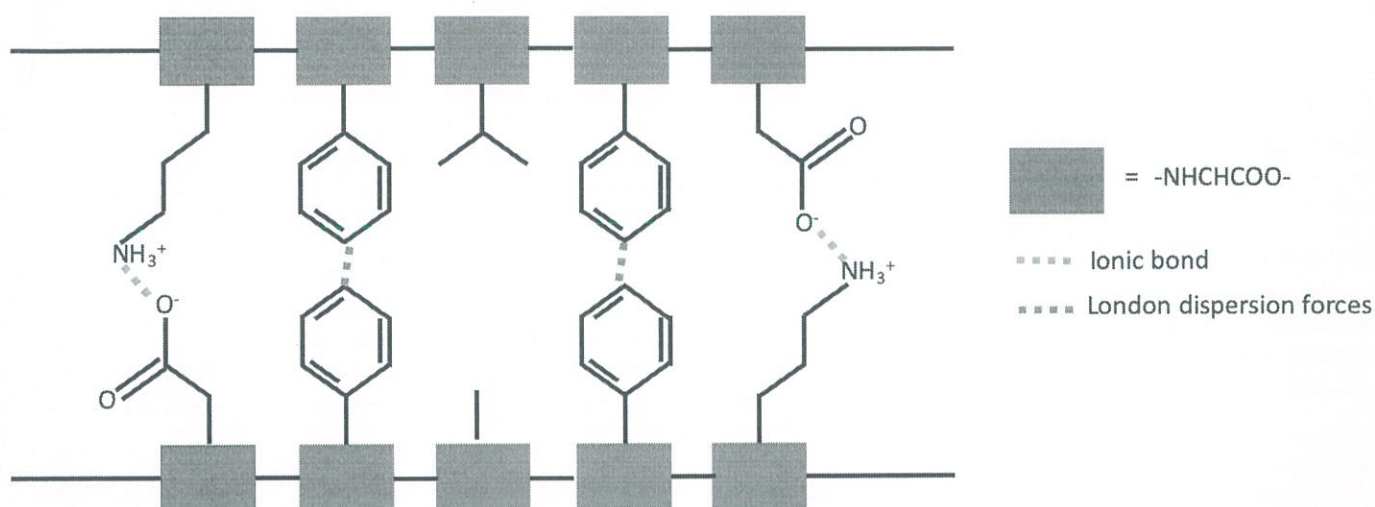
[1]

(b) Write a hydrolysis reaction showing cellulose becoming a monosaccharide



QUESTION 7

The secondary interactions between the side chains of a section of a protein in solution are shown below.



- (a) State and explain the effect of **decreasing** the pH of the solution on the tertiary structure of the protein shown above.

-C(=O)O^- becomes protonated -C(=O)OH
 -NH_3^+ becomes -NH_2
 - ionic bridge becomes ion dipole
 - interactions change
 - shape is altered

[4]

- (b) The protein above is an enzyme. State and explain the effect that decreasing the pH of the solution would have on the function.

will not be able to function as its ability to act depends on its specific 3D shape

[2]

- (c) Draw the structure of the ionic side chains if NaOH was added to the protein solution.



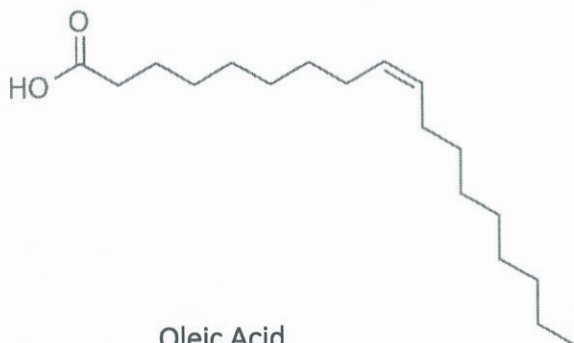
[2]

QUESTION 8

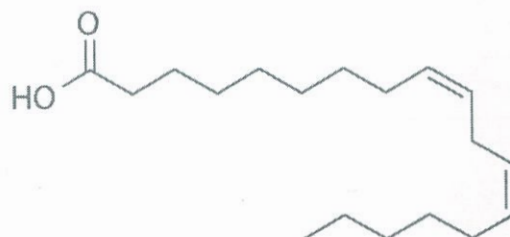
Sunflower oil contains stearic, oleic, and linoleic fatty acids.



Stearic Acid



Oleic Acid



Linoleic Acid

(a) State and explain which one of these fatty acids has the highest boiling point.

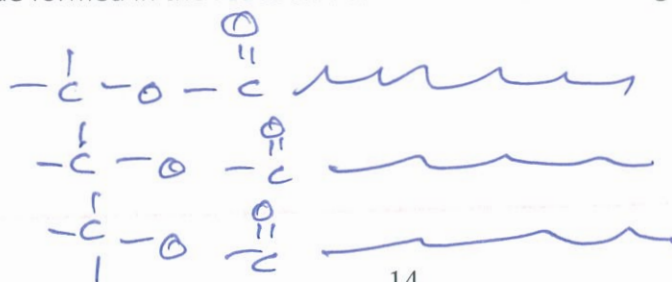
- Stearic
- fully saturated
- molecules pack closely together (same configuration)
- stronger dispersion forces
- more energy needed to overcome forces

[3]

(b) 10.0 g of sunflower oil reacts completely with 123 ml of 0.500 mol L⁻¹ iodine solution.

See back page for iodine number solution

(c) Draw the triglyceride formed in the reaction between stearic acid and glycerol.



[2]

* You may write on this page if you need more space to finish your answers. Make sure to label each answer carefully (e.g. 7(b)(ii)(2) continued).

$$\begin{aligned} \text{I}_2 &\Rightarrow n = c \times v \\ &= 0.5 \times 0.1366 \\ &= 0.0683 \text{ mol} \end{aligned}$$

$$\text{oil} \Rightarrow n = \frac{m}{M_r} = \frac{10 \text{ g}}{876.16 \text{ g mol}^{-1}} = 0.0114 \text{ mol}$$

$$\begin{array}{rcl} \text{I}_2 & : & \text{oil} \\ 0.0683 & : & 0.0114 \\ \hline 0.0114 & & 0.0114 \end{array}$$

$$6 : 1$$

$\therefore$ 6 C=C double bonds